

Imprecision Attenuates Updating: A Characterization*

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Abstract

This paper studies how imprecision in noisy signals attenuates Bayesian updating toward the prior, a property that underlies many comparative-statics results in information economics. This attenuation effect is well-known under a normal prior and normal noise, where less precise signals yield posterior means closer to the prior mean. We show this effect holds for any symmetric, log-concave prior and any symmetric, quasi-concave location experiment, using a newly introduced precision order. Our main result is that this order is characterized by the attenuation effect: the posterior mean under S is closer to the prior mean than under S' for all signal realizations if and only if S is less precise than S' . We relate the precision order to existing information orders and apply it to derive comparative statics for prior precision and overconfidence.

Keywords: signal extraction, reversion to the mean, underreaction, central tendency effect, anchoring and adjustment, cognitive imprecision

JEL Classifications: C11, D83, D84

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1 Introduction

In many economic contexts, agents fail to respond optimally to fundamental variables: firms and households adjust only partially to macroeconomic changes (Sims, 2003), consumers underreact to non-salient taxes (Chetty, Looney, and Kroft, 2009), and numerical estimates are biased toward default values (Tversky and Kahneman, 1974).¹

Such attenuated behavior is often explained through imprecise information. In one version of this explanation, agents base their choices on noisy signals of the fundamentals, where noise arises from observational errors. In a second version, agents observe the fundamentals but the cognitive process that translates fundamentals to actions is noisy, reflecting cognitive constraints. The latter explanation has gained attention in the growing literature on *cognitive imprecision* (Gabaix, 2019; Woodford, 2020; Enke and Graeber, 2023; Enke, Graeber, Oprea, and Yang, 2024), which shows how noisy information processing can account for a wide range of economic phenomena.

These explanations share a core mechanism: noise attenuates Bayesian updating and this compresses behavior toward a default action. This mechanism is usually formalized using the *normal-normal model*, where the agent observes a noisy signal $S = X + \varepsilon$. Here, the state $X \in \mathbb{R}$ is normally distributed, and $\varepsilon \in \mathbb{R}$ is independent, normal noise. The agent’s posterior mean is then compressed toward the prior mean, and more so when the signal is less precise: the posterior mean lies between the signal and the prior mean, and *the posterior mean is closer to the prior mean the larger the variance of ε* .² We refer to this effect as *imprecision attenuates updating*. If behavior is based on the posterior mean, then such attenuation of updating leads to attenuation of behavior. In fact, a key piece of evidence for cognitive imprecision relies on this effect: subjects reporting higher cognitive uncertainty exhibit more attenuated behavioral responses (Enke and Graeber, 2023; Enke, Graeber, Oprea, and Yang, 2024).

Despite its empirical relevance, it remained unknown whether the attenuation effect extends beyond the normal distributions, which rely on strong parametric assumptions. While normality can be justified in specific contexts—e.g., via the central limit theorem or rational inattention with a normal prior and quadratic loss—these cases are limited. A non-parametric class of signal structures that preserves the attenuation effect would offer a more robust foundation for interpreting empirical evidence as a consequence of imprecise information.

To address this gap, we prove the imprecision-attenuates-updating effect holds in all additive-noise models (location experiments) with symmetric, quasi-concave (unimodal) noise under a symmetric, log-concave prior: for any signal realization, the posterior mean moves toward the prior mean as the signal becomes less precise (Theorem 1).

To formalize this result, we introduce a notion of precision, which we call the *precision order*.

¹Beyond economics, psychophysics documents a central-tendency effect: perceptual judgments are biased toward the mean of the stimulus distribution (Hollingworth, 1910; Stevens and Greenbaum, 1966).

²Formally, if $X \sim \mathcal{N}(\mu, \sigma_X^2)$ and $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, then

$$\mathbb{E}[X|S = s] = \frac{\sigma_X^2}{\sigma_X^2 + \sigma_\varepsilon^2} s + \left(1 - \frac{\sigma_X^2}{\sigma_X^2 + \sigma_\varepsilon^2}\right) \mu.$$

For any signal realization s , the larger σ_ε^2 , the closer the posterior mean is to the prior mean.

This order requires the noise distribution of a less precise signal to be farther from zero in the sense of likelihood-ratio domination: $\tilde{S} = X + \tilde{\varepsilon}$ is less precise than $S = X + \varepsilon$ if the likelihood ratio $f_{\tilde{\varepsilon}}(x)/f_{\varepsilon}(x)$ increases as x moves away from 0 (see Figure 1). Concretely, we show that scaling up a log-concave noise term produces a less precise signal under this definition. As a result, the precision order applies to log-concave location-*scale* experiments.³

We show that the precision order is not only sufficient but also *necessary* for the imprecision-attenuates-updating effect. The precision order is thus characterized by this effect. Specifically, if a signal structure S is *not* more precise than another signal structure S' , then there exists a symmetric, log-concave prior and a signal realization where S induces *more* attenuation than S' . This holds even if S dominates S' in the Blackwell or Lehmann orders, which we show do not imply the precision order.

Symmetric, quasi-concave distributions include many common probability distributions beyond the normal, such as logistic, extreme value, and double-exponential distributions. Our focus on this class follows Chambers and Healy (2012), who demonstrate symmetry and quasi-concavity are necessary for the posterior mean to robustly lie between the prior mean and the signal realization. More precisely, they show that if the prior is not symmetric and quasi-concave, one can find a symmetric and quasi-concave noise density (and vice versa) under which the posterior mean does not lie between the prior mean and the signal realization.

Together with Chambers and Healy (2012), this paper shows that the attenuation effect holds under broader conditions than previously recognized. But the effect is not universal. As Chambers and Healy (2012) show, without symmetry and quasi-concavity, agents may overreact—the posterior mean can lie farther from the prior mean than the signal. Moreover, as we show, greater informativeness in the Blackwell sense does not guarantee less attenuation: for some signal realizations, more informative signals can lead to more attenuated updating.

Finally, we leverage Theorem 1 to derive implications of broader economic relevance for signal-extraction problems:

- *Prior Precision*: Our main theorem implies a converse comparative-statics result regarding prior precision (Corollaries 2 and 3): increasing the precision of the prior brings the posterior mean closer to the prior mean for any signal realization.
- *Average Posterior Means*: We extend our analysis to *average* posterior means conditional on the true state, which are relevant when agents acquire conditionally independent signals about a common state. Under symmetry and quasi-concavity, conditional on the true state, the average posterior mean lies between the state and the prior mean (Proposition 1).
- *Comparative Statics for Average Posterior Mean*: Greater (over)confidence in the signal brings the average posterior mean closer to the true state (Proposition 2), whereas greater prior precision brings it closer to the prior mean (Proposition 3).

³Furthermore, we demonstrate that a weaker condition than log-concavity suffices, which holds for many non-log-concave noise distributions (Lemma 1).

Related Literature This paper introduces a new information order and provides comparative-statics results for signal structures with additive independent noise, commonly referred to as location experiments.

Prior work has explored orderings on location experiments related to the value of information. Boll (1955) shows that one location experiment Blackwell-dominates another if and only if the latter’s noise term can be derived by adding an independent noise term to the former’s. This condition is very restrictive, making location experiments comparable only in limited cases. In response, Lehmann (1988) introduced the less restrictive Lehmann order to rank location experiments for *monotone* decision problems. Our precision order is distinct from the Blackwell order, neither implying nor being implied by it. The precision order strengthens the Lehmann order.

Other comparative statics results have been established for location experiments with log-concave noise distributions, which satisfy the monotone likelihood-ratio property. Milgrom (1981) shows that the strict monotone likelihood-ratio property ensures higher signals constitute “good news,” in the sense of producing first-order stochastically dominant posteriors, for any prior.⁴ Conversely, Dawid (1973) examines violations of the “good news”-property under heavy-tailed noise distributions, showing that if the noise is sufficiently heavy-tailed relative to the prior, extreme observations are “rejected,” yielding posteriors that revert to the prior (for further discussion, see O’Hagan and Pericchi, 2012). Together with our results, this highlights log-concave location-scale experiments are a well-behaved class of signal structures that retains key properties of the normal-normal model. Kartik, Lee, and Suen (2021) provides a comparative statics result with economic applications for agents with heterogeneous priors, termed “information validates the prior,” in which a more informative experiment brings another agent’s posterior mean closer to one’s prior mean.

2 Model

We restrict attention to signal structures, where the signal equals the one-dimensional state plus some independent noise, which are called *location experiments*. Formally, the random signal S is a location experiment if it is the sum of the random state X and random noise ε that is independent of X , which we write as $\varepsilon \perp\!\!\!\perp X$. We assume the noise ε has mean 0; that is, the signal is unbiased. If not, one could easily derive an unbiased signal by subtracting the mean of ε from S .

Assumption 1 (Location Experiment).

$$S = X + \varepsilon \tag{1}$$

$$\varepsilon \perp\!\!\!\perp X \tag{2}$$

$$\mathbb{E}[\varepsilon] = 0 \tag{3}$$

Further, we make the technical assumption that the state X and noise ε admit positive, continuously differentiable densities f_X and f_ε with finite absolute first moments. This guarantees finite

⁴See also Chambers and Healy (2011) for a strengthening of Milgrom’s result and Heinsalu (2020) for an extension and economic application of Chambers and Healy (2011).

conditional expectation $\mathbb{E}[X|S = s]$. We call f_X the *prior density* and f_ε the *noise density*.

Assumption 2. *The random state X and noise ε admit positive, continuously differentiable densities f_X, f_ε and have finite absolute first moments.*

Following Chambers and Healy (2012), we assume symmetric and quasi-concave prior and noise densities.⁵ They show that without these assumptions, the posterior mean does not necessarily lie between the prior mean and the signal realization. We believe that our comparative statics result is only interesting when the posterior mean is attenuated in the first place.

Assumption 3. *The prior and noise densities f_X and f_ε are symmetric and quasi-concave.*

In particular, the prior density f_X is symmetric around $\mathbb{E}[X]$ and the noise density f_ε is symmetric around zero by (3).

Chambers and Healy (2012) show the following result, which we include for later reference.

Fact 1 (Chambers and Healy, 2012, Proposition 3). *Under Assumptions 1 to 3, for any signal realization s , the posterior mean $\mathbb{E}[X|S = s]$ lies weakly between the prior mean $\mathbb{E}[X]$ and the signal s . Formally,*

$$\begin{aligned} \forall s \leq \mathbb{E}[X]: \quad & s \leq \mathbb{E}[X|S = s] \leq \mathbb{E}[X], \\ \forall s \geq \mathbb{E}[X]: \quad & s \geq \mathbb{E}[X|S = s] \geq \mathbb{E}[X]. \end{aligned}$$

Throughout the paper, we maintain Assumptions 1 to 3. In particular, all considered prior and noise densities are assumed to be positive, symmetric, and quasi-concave, unless otherwise specified.

3 Precision Order

One challenge to formalizing the idea that imprecision attenuates updating is to find the right order on signal structures. We show that the following new order, which we call the *precision order*, is characteristic of our desired result. Below, we discuss the relation between the precision order and well-known information orders like the Blackwell and Lehmann orders.

We define the precision order for the class of location experiments that satisfy Assumptions 1 to 3. We first define this order on noise terms (or, more generally, on random variables) and then extend it to the associated location experiments.

⁵Quasi-concave densities are also called unimodal densities.

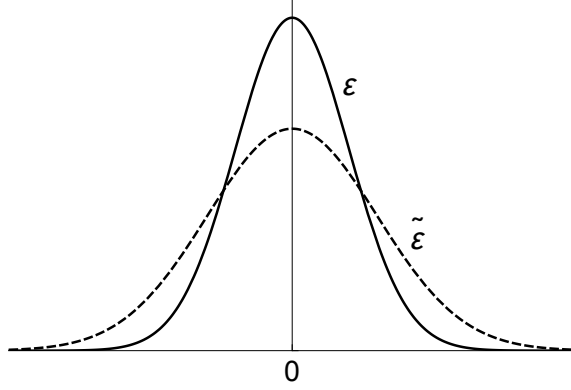


Figure 1: The densities of $\varepsilon \sim \mathcal{N}(0, 1)$ and $\tilde{\varepsilon} \sim \mathcal{N}(0, 1.5^2)$: the former distribution is more precise.

Definition 1 (Precision Order). *Let $\tilde{\varepsilon}$ and ε be random variables with positive, symmetric-around-0, and quasi-concave densities $f_{\tilde{\varepsilon}}$ and f_{ε} , respectively. We say $\tilde{\varepsilon}$ is less precise than ε if the likelihood ratio*

$$\frac{f_{\tilde{\varepsilon}}(x)}{f_{\varepsilon}(x)}$$

is nondecreasing in x for $x > 0$. Further, we say the location experiment $\tilde{S} = X + \tilde{\varepsilon}$ is less precise than location experiment $S = X + \varepsilon$ if $\tilde{\varepsilon}$ is less precise than ε .

Note that by the symmetry of densities, the definition implies the likelihood ratio $f_{\tilde{\varepsilon}}(x)/f_{\varepsilon}(x)$ is weakly decreasing in x for $x < 0$. Thus, the precision order requires that, for positive values, $\tilde{\varepsilon}$ likelihood-ratio dominates ε and, for negative values, $\tilde{\varepsilon}$ is likelihood-ratio dominated by ε . In other words, the less precise location experiment has a noise term further away from 0 in the sense of likelihood-ratio domination. Figure 1 gives an example of two noise distributions ranked by the precision order. The more precise distribution has a higher density at zero, which falls faster as we move away from zero.

A natural question is whether the precision order holds under natural operations on the noise densities. In section 4.1, we show that when we *scale up* a symmetric, log-concave density by a constant $k > 1$, the resulting density is less precise.

How does the precision order relate to existing information orders? The precision order is neither implied by nor implies the Blackwell order. To see why the precision order does not imply by the Blackwell order, consider the latter's restrictive nature for location experiments. As Boll (1955) shows, two location experiments are Blackwell ordered if only if the noise term of one experiment can be obtained from the other's noise term by adding independent noise. Cramér's decomposition theorem further implies that a normal location experiment (i.e., $S = X + \varepsilon$, $\varepsilon \sim \mathcal{N}(\mu, \sigma^2)$) is Blackwell dominated by another location experiment only if the latter is also normal (Cramér, 1936). By contrast, the precision order allows for normal location experiments to be less precise than experiments with non-normal noise. On the other hand, the precision order is not implied by the Blackwell order, as illustrated by the noise term $\tilde{\varepsilon} = \varepsilon + \varepsilon'$, that is the independent sum of a standard Cauchy ε and $\varepsilon' \sim U[-1, 1]$. Computing the density of $\tilde{\varepsilon}$ shows that it is not less precise than ε .

The precision order lies at the intersection of two important orders: the convex order (also called mean-preserving spread order) on noise distributions, and the Lehmann order on location experiments. One can show that if the location experiment $S = X + \varepsilon'$ is less precise than $S = X + \varepsilon$, then noise ε' is a mean-preserving spread of noise ε .⁶ The Lehmann order characterizes what experiments are more valuable in all *monotone* decision problems (Lehmann, 1988; Quah and Strulovici, 2009). This order is defined only for location experiments with log-concave noise, because only these location experiments satisfy the required monotone likelihood-ratio property. One can show for location experiments with log-concave noise, the precision order implies the Lehmann order.⁷

4 Results

Our main result establishes that *imprecision attenuates updating* if and only if location experiments are ranked by the precision order, under the additional assumption of a log-concave prior. Recall that throughout the paper, all considered location experiments are assumed to satisfy Assumptions 1 to 3.

Theorem 1 (Imprecision Attenuates Updating). *Let the prior density be log-concave. Consider two location experiments $S = X + \varepsilon$ and $\tilde{S} = X + \tilde{\varepsilon}$. For every signal realization, the posterior mean under $\tilde{S}$ is closer to the prior mean than is the posterior mean under S , if and only if $\tilde{S}$ is less precise than S . Formally,*

$$\begin{aligned} \forall s \leq \mathbb{E}[X]: \quad & \mathbb{E}[X|S = s] \leq \mathbb{E}[X|\tilde{S} = s] \leq \mathbb{E}[X] \\ \forall s \geq \mathbb{E}[X]: \quad & \mathbb{E}[X|S = s] \geq \mathbb{E}[X|\tilde{S} = s] \geq \mathbb{E}[X] \end{aligned} \tag{4}$$

if and only if $\tilde{S}$ is less precise than S .

As an illustration, consider two agents who observe the same signal, such as an economic forecast, but differ in their assessment of the signal precision. The agent who assumes greater precision forms a posterior mean closer to the signal, for any the signal realization.

The proof, relegated to the Appendix, demonstrates necessity and sufficiency of the precision order. Necessity is the easier direction and shown by constructing a counterexample when two location experiments are not precision-ordered. Sufficiency is more involved. We can suppose without loss that the signal realization s is larger than the prior mean. To the right of s , the more precise

⁶If S' is less precise than S , the CDFs of ε' and ε cross once. Together with ε' and ε having equal means, this implies one distribution majorizes the other; see also Stiglitz (1974). Note that a mean-preserving spread is less demanding than the Blackwell order on location experiments, which requires that the additional noise is independent.

⁷On such location experiments, the Lehmann order coincides with the dispersive order on the noise distributions, which requires that any two quantiles are weakly further apart under the more dispersed noise distribution. Formally, let ε and ε' have CDFs F and G , respectively, and let F^{-1} and G^{-1} denote here the right-continuous inverses. ε is smaller in the dispersive order than ε' if $\forall 0 < \alpha \leq \beta < 1 : F^{-1}(\beta) - F^{-1}(\alpha) \leq G^{-1}(\beta) - G^{-1}(\alpha)$. When we truncate the noise to positive values, the precision order implies the likelihood-ratio order. As is known, the likelihood-ratio order implies the hazard-rate order (e.g., Shaked and Shanthikumar, 2007, Theorem 1.C.1). The hazard-rate order together with densities being log-concave implies the dispersive order (Bagai and Kochar, 1986, Theorem 1). One can easily see that if the dispersive order holds for all $1/2 \leq \alpha \leq \beta < 1$, then it holds for all $0 < \alpha \leq \beta < 1$.

signal results in a likelihood-ratio dominated posterior, lowering the posterior mean. To the left of the s , the more precise signal results in a likelihood-ratio dominant posterior, raising the posterior mean. The proof establishes that the latter effect dominates, using a sequence of inequalities and bounds, leveraging symmetry, quasi-, and log-concavity assumptions.

For this as well as the following results, an analogous *strict* version of the result holds. We can require that the noise term is replaced by a *strictly* less precise noise in the sense that the likelihood ratio in Definition 1 is strictly decreasing. Then, for signal realizations distinct from the prior mean, the posterior mean is strictly closer to s .

Next, we show that our precision order arises endogenously in *location-scale experiments* with log-concave, and many non-log-concave, noise densities.

4.1 Location-Scale Experiments

We introduce to our location experiment a *scale parameter* $\sigma \in \mathbb{R}_{\geq 0}$ that scales the noise term, such that

$$S_\sigma = X + \sigma \cdot \varepsilon. \quad (5)$$

We show comparative statics on the scale parameter σ .

Lemma 1. *Let f_ε be symmetric around 0. Then $\forall \sigma' > \sigma > 0$, $\sigma' \varepsilon$ is less precise than $\sigma \varepsilon$ if and only if $f_\varepsilon(e^x)$ is log-concave.*

The condition in 1 is easy to verify and holds broadly. If f_ε is symmetric around 0 and log-concave, then $f_\varepsilon(e^x)$ is log-concave.⁸ Thus, all location-scale experiments with symmetric, log-concave error densities are covered, including the normal, logistic, double-exponential, and uniform.⁹ The condition also holds beyond log-concavity: in Appendix 6.2, we verify it for the Student- t (including Cauchy) and double-Pareto distributions.

Together, Theorem 1 and Lemma 1 imply the following result. By Lemma 1, the result still holds if we weaken the assumption that the noise density f_ε is log-concave to $\log f_\varepsilon(e^x)$ being concave.

Corollary 1. *Let the prior and noise densities be log-concave. The posterior mean is weakly closer to the prior mean under a larger scale parameter, for any signal realization s . Formally, if $\tilde{\sigma} > \sigma > 0$, then*

$$\begin{aligned} \forall s \leq \mathbb{E}[X]: \quad & \mathbb{E}[X|S_\sigma = s] \leq \mathbb{E}[X|S_{\tilde{\sigma}} = s] \leq \mathbb{E}[X], \\ \forall s \geq \mathbb{E}[X]: \quad & \mathbb{E}[X|S_\sigma = s] \geq \mathbb{E}[X|S_{\tilde{\sigma}} = s] \geq \mathbb{E}[X]. \end{aligned}$$

The next section shows comparative statics results for changing the prior instead of the noise, exploiting a symmetry in location experiments.

⁸To see this, note that $(\log f \circ \exp)''(x) = e^x(\log f)'(e^x) + e^{2x}(\log f)''(e^x) < 0$, since $(\log f)'' < 0$ by log-concavity and $(\log f)'(x) < 0$ for $x > 0$ by symmetry and log-concavity.

⁹The uniform distribution has bounded support and so violates Assumption 2. However, any uniform distribution can be approximated by a log-concave distribution with full support (see the proof of Lemma 2), so our results extend to uniform errors.

4.2 Comparative Statics on the Prior

Location experiments exhibit a useful duality between the prior and the noise density: the signal $S = X + \varepsilon$ can be viewed as a location experiment about ε with noise X . Applying Theorem 1 with the roles of state and noise interchanged yields a dual result for making the prior more precise.

Corollary 2. *Let the noise density be log-concave. The posterior mean is weakly closer to the prior mean under a more precise prior, for any signal realization s . Formally, if $\mathbb{E}[X] = \mathbb{E}[\tilde{X}]$ and $\tilde{X}$ is more precise than X , then*

$$\begin{aligned} \forall s \leq \mathbb{E}[X]: \quad & \mathbb{E}[X|X + \varepsilon = s] \leq \mathbb{E}[\tilde{X}|\tilde{X} + \varepsilon = s] \leq \mathbb{E}[X], \\ \forall s \geq \mathbb{E}[X]: \quad & \mathbb{E}[X|X + \varepsilon = s] \geq \mathbb{E}[\tilde{X}|\tilde{X} + \varepsilon = s] \geq \mathbb{E}[X]. \end{aligned}$$

To illustrate Corollary 2, suppose that two agents observe the same signal about the state but one agent has a more precise prior (but with the same mean). Corollary 2 implies that the agent with the more precise prior has a posterior mean that is closer to the prior mean, for any signal realization.

By the same argument, an analogous dual result to Corollary 1 holds for scaling the prior instead of the noise density. If X has density $f_X(x)$, then kX has density $1/k f_X(x/k)$.

Corollary 3. *Let the prior and noise densities be log-concave. For any signal realization s , the posterior mean is weakly closer to the prior mean if we scale down the prior. Formally, if $0 < \tilde{k} < k$ and we normalize the prior mean to zero, $\mathbb{E}[X] = 0$, then*

$$\begin{aligned} \forall s \leq \mathbb{E}[X]: \quad & \mathbb{E}[kX|kX + \varepsilon = s] \leq \mathbb{E}[\tilde{k}X|\tilde{k}X + \varepsilon = s] \leq \mathbb{E}[X], \\ \forall s \geq \mathbb{E}[X]: \quad & \mathbb{E}[kX|kX + \varepsilon = s] \geq \mathbb{E}[\tilde{k}X|\tilde{k}X + \varepsilon = s] \geq \mathbb{E}[X]. \end{aligned}$$

4.3 Average Posterior Means

Our previous comparative statics results hold for any signal realization and thus speaks to situations where agents observe the same signal. However, in many situations, agents observe *distinct* signals from the same signal structure, that is their signal realization are independent conditional on the state. Further, often we do not observe agent's signal realizations but only the *average* posterior mean, such as when we observe only aggregate behavior from a population of individuals. What can be said about comparative statics with respect to the average posterior means given some true state?

Before we prove comparative statics results, we show that the *average* posterior mean necessarily lies between the state X and the prior mean $\mathbb{E}[X]$, extending Fact 1.

Proposition 1. *For any state x , the conditional average posterior mean $\mathbb{E}[\mathbb{E}[X|S]|X = x]$ lies weakly between the state x and the prior mean $\mathbb{E}[X]$. Formally,*

$$\begin{aligned} \forall x \leq \mathbb{E}[X]: \quad & x \leq \mathbb{E}[\mathbb{E}[X|S]|X = x] \leq \mathbb{E}[X], \\ \forall x \geq \mathbb{E}[X]: \quad & x \geq \mathbb{E}[\mathbb{E}[X|S]|X = x] \geq \mathbb{E}[X]. \end{aligned}$$

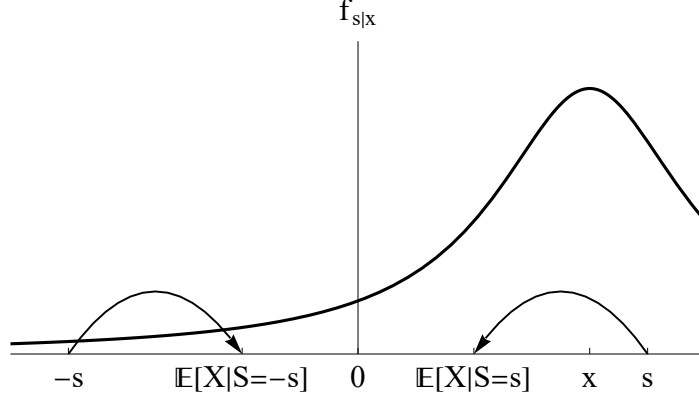


Figure 2: Illustrating the proof of Proposition 1.

The proof is illustrated using Figure 2. Let the state be greater than expected and normalize the expectation to 0, $x > \mathbb{E}[X] = 0$. Conditional on $X = x$, the distribution of the signal S is symmetric around x so its expectation equals x . Pushing forward the conditional distribution of the signal through the conditional-expectation function $\mathbb{E}[X|S]$ moves the distribution closer to $\mathbb{E}[X] = 0$ as indicated by the arrows. Given our assumptions, the conditional expectation is antisymmetric and $s > 0$ has a higher likelihood than $-s < 0$. Thus, the overall effect on the average posterior mean $\mathbb{E}[\mathbb{E}[X|S]|X = x]$ is to push it below $\mathbb{E}[S|X] = x$. Further, because the density of s is larger than the density of $-s$, integrating over all s leads to a positive expectation, so $\mathbb{E}[X] < \mathbb{E}[\mathbb{E}[X|S]|X = x]$.

Overconfidence First, we prove a comparative statics result for overconfidence in the signal. We consider two agents, A and B , that face the same objective signal structure $S = X + \varepsilon$ but update differently because they have different confidence in the signal. That is, agent $i \in \{A, B\}$ forms their conditional expectation $\mathbb{E}_i[X|S = s]$ as if $S = X + \varepsilon_i$, $\mathbb{E}_i[X|S = s] := \mathbb{E}[X|X + \varepsilon_i = s]$. Especially empirically relevant is the case *overconfidence* in the signal, also called *overprecision*, which is pervasive (Moore and Healy, 2008). We define a relative notion of (over)confidence by generalizing the definition in Ortoleva and Snowberg (2015), which is based on the normal-normal model, using our Definition 1.

Definition 2. *A is more confident than B in the signal if ε_A is more precise than ε_B .*

Using this definition, we prove the following comparative statics result.

Proposition 2. *Let the prior density be log-concave and A be more confident in the signal than B. Conditional on any state x , the average posterior mean of A is weakly closer to the state than the average posterior mean of B. Formally,*

$$\begin{aligned} \forall x \leq \mathbb{E}[X]: \quad & x \leq \mathbb{E}[\mathbb{E}_A[X|S]|X = x] \leq \mathbb{E}[\mathbb{E}_B[X|S]|X = x] \leq \mathbb{E}[X], \\ \forall x \geq \mathbb{E}[X]: \quad & x \geq \mathbb{E}[\mathbb{E}_A[X|S]|X = x] \geq \mathbb{E}[\mathbb{E}_B[X|S]|X = x] \geq \mathbb{E}[X]. \end{aligned}$$

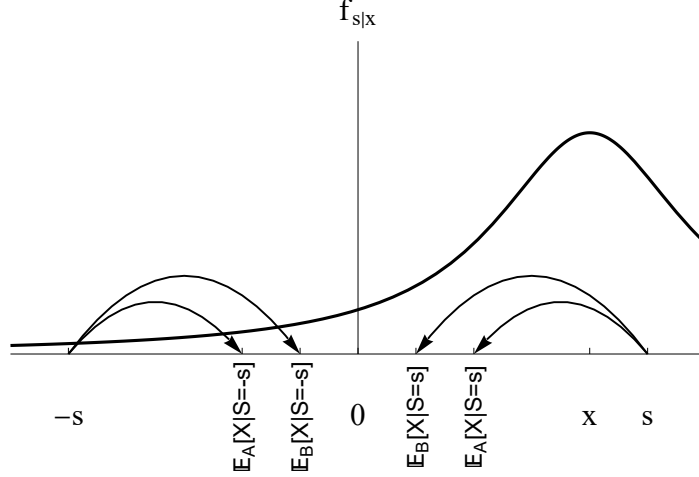


Figure 3: Proof of Proposition 2

The result states that the average posterior mean of the more confident agent is closer to the true state X and further away from the prior mean $\mathbb{E}[X]$.

We sketch the proof using Figure 3. We suppose again that $x > \mathbb{E}[X] = 0$. Because A is more confident in the signal, their posterior mean is closer to the signal for signal realization s as well as $-s$. Thus, the posterior mean of A is sometimes greater and sometimes less than the posterior mean of B . However, for $x > 0$, because $s > 0$ is more likely than $-s$, the positive effect dominates and the average posterior mean of A is greater than the one of B .

Prior Precision Second, we show comparative statics with respect to the prior precision. Given some state x , consider two agents, A and B , with symmetric and quasi-concave priors that have the same mean, where agent A 's prior is less precise than B 's. Then, using the proof of Proposition 2 and Corollary 2, we obtain the following result.

Proposition 3. *Let the noise density be log-concave and A 's prior be less precise than B 's. Conditional on any state x , the average posterior mean of A is weakly closer to the state than the average posterior mean of B . Formally,*

$$\begin{aligned} \forall x \leq \mathbb{E}[X]: \quad & x \leq \mathbb{E}[\mathbb{E}_A[X|S]|X=x] \leq \mathbb{E}[\mathbb{E}_B[X|S]|X=x] \leq \mathbb{E}[X], \\ \forall x \geq \mathbb{E}[X]: \quad & x \geq \mathbb{E}[\mathbb{E}_A[X|S]|X=x] \geq \mathbb{E}[\mathbb{E}_B[X|S]|X=x] \geq \mathbb{E}[X]. \end{aligned}$$

Prior precision has the opposite effect from confidence in the signal. A more precise prior leads to an average posterior mean closer to the prior mean and further from the signal.

5 Conclusion

In this paper, we extended the attenuation effect of imprecise information beyond the normal-normal model. By introducing a new order of precision, based on likelihood-ratio dominance, we demon-

strated that imprecision attenuates expectations toward the prior mean across a broad class of distributions commonly used in economic modeling. This generalization provides a more robust theoretical foundation for interpreting empirical observations of attenuated behavior as resulting from imprecision, whether observational or cognitive.

Our findings also have broader implications for signal-extraction problems. We established comparative statics results regarding prior precision, showing that increased prior precision shifts the posterior mean toward the prior mean, for any given signal realization. Additionally, we analyzed how overconfidence and prior precision affect the *average* posterior mean. Perhaps surprisingly, our results show that the average posterior mean of overconfident agents is closer to the truth while it is further away for agents with more precise priors.

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6 Appendix: Proofs

6.1 Theorem 1

Proof. We, first, show necessity and, second, sufficiency of the precision order for the ordering requirement (4).

Lemma 2 (Necessity). *If $\tilde{\varepsilon}$ is not less precise than ε , then there exist a symmetric, log-concave prior and a signal realization $s \in \mathbb{R}$ such that the ordering requirement (4) of Theorem 1 is violated.*

Proof. We first find a uniform prior and a signal realization s such that (4) is violated. Then, we approximate the uniform prior with a symmetric, log-concave prior density that is positive everywhere

First, if $\tilde{\varepsilon}$ is not less precise than ε , then the likelihood-ratio $f_{\tilde{\varepsilon}}(x)/f_{\varepsilon}(x)$ has a strictly negative derivative for some $x = s > 0$.¹⁰ By continuous differentiability, $f_{\tilde{\varepsilon}}(x)/f_{\varepsilon}(x)$ has a strictly negative derivative on some interval $[s - \delta, s + \delta]$ around s .

Consider the uniform prior on $[-\delta, \delta]$, $f_X^U(x) = 1/(2\delta)$. Let f be the posterior density upon observing $X + \tilde{\varepsilon} = s$ and g be the posterior density upon observing $X + \varepsilon = s$. The posterior density

¹⁰If both noise densities f_{ε} and $f_{\tilde{\varepsilon}}$ are continuously differentiable and positive, then the likelihood ratio $f_{\tilde{\varepsilon}}/f_{\varepsilon}$ is continuously differentiable.

$f(x)$ is proportional to the product $f_X^U(x)f_{\tilde{\varepsilon}}(s-x)$ and the posterior density $g(x)$ is proportional to the product $f_X^U(x)f_{\varepsilon}(s-x)$, so the ratio $f(x)/g(x)$ of the posterior densities is proportional to

$$\frac{f_{\tilde{\varepsilon}}(s-x)}{f_{\varepsilon}(s-x)}.$$

By the above, this likelihood ratio is strictly increasing in x on the support $[-\delta, \delta]$ of the prior, so the posterior under signal $\tilde{S} = s$ strictly likelihood-ratio dominates the posterior under signal $S = s$. Hence, the posterior mean is strictly greater under the less precise signal, $\mathbb{E}[X|X + \tilde{\varepsilon} = s] > \mathbb{E}[X|X + \varepsilon = s]$, despite $s > 0 = \mathbb{E}[X]$, violating (4).

Second, we approximate the uniform prior $[-\delta, \delta]$ with $\delta > 0$ with a symmetric, log-concave prior, such that the posterior mean converges (pointwise) for any signal realization s . For any $d > 0$, define the prior density $f_X^d(x)$ via

$$\log f_X^d(x) = \begin{cases} c, & \text{if } x \in [-\delta, \delta] \\ c - d(x - \delta)^2, & \text{if } x > \delta \\ c - d(x - (-\delta))^2, & \text{if } x < -\delta. \end{cases}$$

where c is the unique constant such that f_X integrates to 1. The density f_X is symmetric around 0, log-concave, and integrable. As we let d go to $+\infty$, the density f_X^d pointwise converges to the uniform density f_X^U . Using Lebesgue dominated convergence theorem, we show that as we let d go to $+\infty$, the posterior mean under signal realization s converges:

$$\frac{\int x f_X^d(x) f_{\varepsilon}(s-x) dx}{\int f_X^d(x) f_{\varepsilon}(s-x) dx} \xrightarrow{d \rightarrow \infty} \frac{\int x f_X^U(x) f_{\varepsilon}(s-x) dx}{\int f_X^U(x) f_{\varepsilon}(s-x) dx}.$$

To show that the numerator converges, note that by $f_X^d(x) < 1/(2\delta)$, the integrand is bounded by $x f_{\varepsilon}(s-x)/(2\delta)$, which is integrable by ε having a finite first absolute moment. Thus, by Lebesgue dominated convergence theorem, the integral converges to $\int x f_X^U(x) f_{\varepsilon}(s-x) dx$.

To show that the denominator converges, note that for any $d > 0$, $f_X^d(x) < 1/(2\delta)$. Thus, the integrand is bounded by $f_{\varepsilon}(s-x)/(2\delta)$, which is integrable. By the Lebesgue dominated convergence theorem, the integral converges to $\int f_X^U(x) f_{\varepsilon}(s-x) dx$. Because the denominator is strictly positive, the ratio converges.

By $\mathbb{E}[X|X + \tilde{\varepsilon} = s] > \mathbb{E}[X|X + \varepsilon = s]$ under uniform prior f_X^U , we can choose a d large enough such that the same ordering requirement holds under prior f_X^d . $\square$

The rest of the proof concerns the harder direction, that is, showing that the precision order is *sufficient* for (4). Without loss of generality, the signal realization is zero, $s = 0$. If $\mathbb{E}[X] = 0$, then the posterior mean is zero by Fact 1 and we are done. Assume $\mathbb{E}[X] < 0$ (the case $\mathbb{E}[X] > 0$ is analogous). We prove that for strictly more precise noise, the posterior mean becomes strictly closer to 0. Let $\tilde{\varepsilon}$ be less precise than ε , and $f_{\tilde{\varepsilon}}$ and f_{ε} denote their respective densities. Further, let f denote the posterior density under noise $\tilde{\varepsilon}$ after observing signal $s = 0$ and g analogously under noise

ε .

The posterior mean is

$$\mathbb{E}[X|X + \varepsilon = 0] = \int_{-\infty}^0 xg(x)dx + \int_0^{\infty} xg(x)dx = \int_0^{\infty} -x(g(-x) - g(x))dx,$$

and analogously with density f instead of g for the signal with greater noise. The proof revolves around showing that the following integral is positive:

$$\mathbb{E}[X|X + \varepsilon = 0] - \mathbb{E}[X|X + \tilde{\varepsilon} = 0] = \int_0^{\infty} -x[(g(-x) - g(x)) - (f(-x) - f(x))]dx \quad (6)$$

First, we prove the following result regarding the integrand of (6), which uses that $\frac{f_{\varepsilon}(x)}{f_{\tilde{\varepsilon}}(x)}$ is strictly decreasing for $x > 0$.

Lemma 3. *There is some $c > 0$ such that the integrand of (6) is strictly negative for $x \in [0, c)$ and strictly positive for $x \in (c, \infty)$.*

Proof. Again, the density $f(x)$ is proportional to $f_X(x)f_{\tilde{\varepsilon}}(x)$ and $g(x)$ to $f_X(x)f_{\varepsilon}(x)$. Thus, there is some factor $C > 0$ such that

$$g(-x) - g(x) = C \frac{f_{\varepsilon}(-x)}{f_{\tilde{\varepsilon}}(-x)} f(-x) + C \frac{f_{\varepsilon}(x)}{f_{\tilde{\varepsilon}}(x)} f(x) = C \frac{f_{\varepsilon}(x)}{f_{\tilde{\varepsilon}}(x)} (f(-x) - f(x)),$$

where we have used the symmetry of the noise densities. Thus, the integrand of (6) is negative if, and only if, $C \frac{f_{\varepsilon}(x)}{f_{\tilde{\varepsilon}}(x)} > 1$.

The ratio $\frac{f_{\varepsilon}(x)}{f_{\tilde{\varepsilon}}(x)}$ is strictly decreasing for $x > 0$ by assumption. As both are densities that integrate to 1, the ratio must cross $1/C$ and by strictly decreasing ratio, the crossing point must be unique up to sign. Let c be the unique positive x at which $C \frac{f_{\varepsilon}(x)}{f_{\tilde{\varepsilon}}(x)} = 1$. Then, for $x \in [0, c)$ we have $C \frac{f_{\varepsilon}(x)}{f_{\tilde{\varepsilon}}(x)} > 1$ and for $x \in (c, \infty)$ we have $C \frac{f_{\varepsilon}(x)}{f_{\tilde{\varepsilon}}(x)} < 1$. $\square$

Without loss, we can rescale the space, so that $c = 1$. Using Lemma 3, we have that

$$\begin{aligned} & \int_0^{\infty} -x[(g(-x) - g(x)) - (f(-x) - f(x))]dx \\ & > \\ & \int_0^1 -1 \cdot [(g(-x) - g(x)) - (f(-x) - f(x))]dx + \int_1^{\infty} -1 \cdot [(g(-x) - g(x)) - (f(-x) - f(x))]dx \\ & = \\ & (-G_2 + G_3 - G_1 + G_4) - (-F_2 + F_3 - F_1 + F_4) \end{aligned} \quad (7)$$

where F_1 through F_4 are the probabilities of according to f on four mutually exclusive and exhaustive

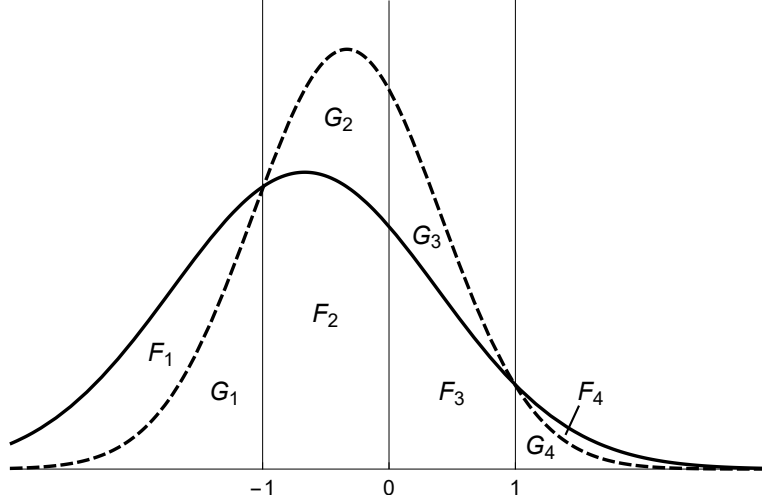


Figure 4: The posteriors f (bold) and g (dashed) after observing $X + \tilde{\varepsilon} = 0$ and $X + \varepsilon = 0$, respectively.

intervals

$$\begin{aligned}
F_1 &:= \int_1^\infty f(-x)dx = \int_{-\infty}^{-1} f(x)dx \\
F_2 &:= \int_0^1 f(-x)dx = \int_{-1}^0 f(x)dx \\
F_3 &:= \int_0^1 f(x)dx \\
F_4 &:= \int_1^\infty f(x)dx
\end{aligned}$$

so $F_1, F_2, F_3, F_4 > 0$ and $F_1 + F_2 + F_3 + F_4 = 1$. G_1 through G_4 are defined analogously. We thus need to show that

$$-G_2 + G_3 - G_1 + G_4 > -F_2 + F_3 - F_1 + F_4. \quad (8)$$

The proof proceeds as follows. As the Figure 4 depicts, G_2 and G_3 are larger than F_2 and F_3 , respectively, and G_1 and G_4 are smaller than F_1 and F_4 , respectively. We show in Lemma 7 that (8) would hold if G_2 and G_3 were larger than F_2 and F_3 , each, by the same factor and similarly for G_1 and G_4 . Lemma 4 to 6 argue that in fact these ratios are not the same and thus $-G_2 + G_3 - G_1 + G_4$ is even larger, proving (8).

Before that, we prove the following lemma, which uses the log-concavity and symmetry of the prior density as well as the symmetry of the noise density.

Lemma 4. *The posterior probability-ratio $\frac{f(-x)}{f(x)}$ is strictly increasing in x .*

Proof. By Bayes' law, f is proportional to the product $f_X(x)f_{\tilde{\varepsilon}}(s-x) = f_X(x)f_{\tilde{\varepsilon}}(x)$. The ratio is strictly increasing if its logarithm, which is as follows, is strictly increasing in x . Using the symmetry

of f_{ε} and f_X we obtain

$$\begin{aligned}\log\left(\frac{f(-x)}{f(x)}\right) &= \log\left(\frac{f_X(-x)f_{\varepsilon}(-x)}{f_X(x)f_{\varepsilon}(x)}\right) = \log\left(\frac{f_X(-x)}{f_X(x)}\right) \\ &= \log f_X(-x) - \log f_X(x) = \log f_X(x + 2\mathbb{E}[X]) - \log f_X(x),\end{aligned}$$

where we have used that $f_X(x)$ is symmetric around $\mathbb{E}[X] < 0$, so

$$\log f_X(-x) = \log f_X(\mathbb{E}[X] + (-x - \mathbb{E}[X])) = \log f_X(\mathbb{E}[X] - (-x - \mathbb{E}[X])) = \log f_X(x + 2\mathbb{E}[X]).$$

By strict concavity of $\log f_X$ and $\mathbb{E}[X] < 0$, the difference $\log f_X(x + 2\mathbb{E}[X]) - \log f_X(x)$ is strictly increasing. $\square$

Using Lemma 4, we prove two lemmas regarding ratios of the terms in (7).

Lemma 5. $\frac{F_1}{F_4} \geq \frac{F_2}{F_3}$.

Proof. We have that

$$\frac{F_1}{F_4} = \frac{\int_1^\infty f(-x)dx}{\int_1^\infty f(x)dx} = \frac{\int_1^\infty \frac{f(-x)}{f(x)}f(x)dx}{\int_1^\infty f(x)dx} \geq \frac{\int_0^1 \frac{f(-x)}{f(x)}f(x)dx}{\int_0^1 f(x)dx} = \frac{\int_0^1 f(-x)dx}{\int_0^1 f(x)dx} = \frac{F_2}{F_3}.$$

The inequality holds because by Lemma 4. The two inner terms are the expectation of $\frac{f(-x)}{f(x)}$ with respect to the posterior distribution f conditional on the domain $[1, \infty)$ and $[0, 1]$, respectively. The former distribution first-order stochastically dominates the latter, thus the inequality follows from $\frac{f(-x)}{f(x)}$ being strictly increasing in x for $x > 0$. $\square$

The following lemma uses that $\frac{f_{\varepsilon}(x)}{f_{\varepsilon}(x)}$ is decreasing for $x > 0$.

Lemma 6. $1 < \frac{G_2}{F_2} < \frac{G_3}{F_3}$ and $\frac{G_1}{F_1} < \frac{G_4}{F_4} < 1$.

Proof. Using $g(x) = \frac{f_{\varepsilon}(x)}{f_{\varepsilon}(x)}f(x)C$, where C is the ratio of the integration constants, and $\frac{f_{\varepsilon}(-x)}{f_{\varepsilon}(-x)} = \frac{f_{\varepsilon}(x)}{f_{\varepsilon}(x)}$ (by symmetry), we have

$$\begin{aligned}\frac{G_2}{F_2} &= \frac{\int_0^1 \frac{g(-x)}{f(-x)}f(-x)Cdx}{\int_0^1 f(-x)dx} = \frac{\int_0^1 \frac{f_{\varepsilon}(x)}{f_{\varepsilon}(x)}Cf(-x)dx}{\int_0^1 f(-x)dx} \\ \frac{G_3}{F_3} &= \frac{\int_0^1 \frac{g(x)}{f(x)}f(x)dx}{\int_0^1 f(x)dx} = \frac{\int_0^1 \frac{f_{\varepsilon}(x)}{f_{\varepsilon}(x)}Cf(x)dx}{\int_0^1 f(x)dx}.\end{aligned}$$

Thus, both ratios are expectations of the function $\frac{f_{\varepsilon}(x)}{f_{\varepsilon}(x)}$ over the interval $[0, 1]$ multiplied by C but with densities $f(-x)/(\int_0^1 f(-x)dx)$ and $f(x)/(\int_0^1 f(x)dx)$, respectively. Because $\frac{f(-x)}{f(x)}$ is increasing in $x > 0$ by Lemma 4, the former density likelihood-ratio dominates the latter. This implies first-order stochastic dominance, which in turn implies a strictly smaller expectation since $\frac{f_{\varepsilon}(x)}{f_{\varepsilon}(x)}$ is a strictly decreasing function by assumption. Thus, $\frac{G_2}{F_2} < \frac{G_3}{F_3}$.

Moreover, by Lemma 3 and having normalized $c = 1, \frac{f_\varepsilon(x)}{f_{\tilde{\varepsilon}}(x)}C > 1$ in the interval $[0, 1)$. Hence, $\frac{G_2}{F_2}$ and $\frac{G_3}{F_3}$, which are expectations of this ratio, are strictly greater than 1.

The proof of $\frac{G_1}{F_1} < \frac{G_4}{F_4} < 1$ is analogous, but with expectation over the domain $[1, \infty)$. $\square$

Define $\tilde{G}_2 = kF_2$ and $\tilde{G}_3 = kF_3$ with $k > 1$, and $\tilde{G}_1 = lF_1$ and $\tilde{G}_4 = lF_4$ with $l < 1$, such that $\tilde{G}_2 + \tilde{G}_3 = G_2 + G_3$ and $\tilde{G}_1 + \tilde{G}_4 = G_1 + G_4$. By Lemma 6, $\frac{G_3}{F_3} > \frac{\tilde{G}_3}{F_3} = \frac{G_2 + G_3}{F_2 + F_3} = \frac{\tilde{G}_2}{F_2} > \frac{G_2}{F_2}$, so $Q_3 > \tilde{Q}_3$ and $\tilde{Q}_2 > Q_2$, implying $-G_2 + G_3 > -\tilde{G}_2 + \tilde{G}_3$. Analogously, $-G_1 + G_4 > -\tilde{G}_1 + \tilde{G}_4$.

$$-G_2 + G_3 - G_1 + G_4 > -\tilde{G}_2 + \tilde{G}_3 - \tilde{G}_1 + \tilde{G}_4. \quad (9)$$

Finally, the following lemma concludes the proof.

Lemma 7. $-\tilde{G}_2 + \tilde{G}_3 - \tilde{G}_1 + \tilde{G}_4 > -F_2 + F_3 - F_1 + F_4$.

Proof. Define $r := \frac{F_2}{F_3} > 0$ and $R := \frac{F_1}{F_4} > 0$ where $R > r$ by Lemma 5, and $P := F_2 + F_3$ and $p := F_1 + F_4$.

We have $\tilde{G}_1 + \tilde{G}_2 + \tilde{G}_3 + \tilde{G}_4 = G_1 + G_2 + G_3 + G_4 = F_1 + F_2 + F_3 + F_4 = 1$. Thus, we can define $\Delta := (\tilde{G}_2 + \tilde{G}_3) - (F_2 + F_3) = (F_1 + F_4) - (\tilde{G}_1 + \tilde{G}_4)$ with $\Delta > 0$ as well as $kP = P + \Delta$ and $lp = p - \Delta$. From $P = F_2 + F_3$ and $r = F_2/F_3$, it follows that $-F_2 + F_3 = P(\frac{1}{1+r} - \frac{r}{1+r}) = -P\frac{r-1}{r+1}$ and analogously $-F_1 + F_4 = -p\frac{R-1}{R+1}$. So, $(-\tilde{G}_2 + \tilde{G}_3) - (-F_2 + F_3) = -\Delta\frac{r-1}{r+1}$ and $(-\tilde{G}_1 + \tilde{G}_4) - (-F_1 + F_4) = \Delta\frac{R-1}{R+1}$. Note that $\frac{d}{dx} \frac{x-1}{x+1} = \frac{2}{(x+1)^2} > 0$ for $x > 0$. Then, by $R > r$, $\frac{R-1}{R+1} > \frac{r-1}{r+1}$, so $-\tilde{G}_2 + \tilde{G}_3 - \tilde{G}_1 + \tilde{G}_4 > -F_2 + F_3 - F_1 + F_4$. $\square$

By (9) and Lemma 7, we obtain $-G_2 + G_3 - G_1 + G_4 > -F_2 + F_3 - F_1 + F_4$. $\square$

6.2 Lemma 1

Proof. By our definition, the symmetric around 0 random variable ε is more precise than the symmetric around 0 random variable $\tilde{\varepsilon}$ if $[\varepsilon|\varepsilon > 0]$ is smaller in the likelihood ratio order than $[\tilde{\varepsilon}|\tilde{\varepsilon} > 0]$. For a non-negative continuous random variables X , aX is smaller in the likelihood ratio order than X for all $0 < a < 1$ if, and only if, $\log f_\varepsilon(e^x)$ is concave (Hu, Nanda, Xie, and Zhu, 2004, Remark 5 (iii)). Applying this to $X = [\varepsilon|\varepsilon > 0]$ yields the result. $\square$

The main text gives several examples of commonly encountered symmetric, log-concave distributions. Below, we prove for that several symmetric distributions that are not log-concave that $\log f_\varepsilon(e^x)$ is nevertheless concave.

Non-log-concave examples The *Student-t* distribution with parameter $\nu > 0$, which includes as a special case the Cauchy distribution, gives

$$\begin{aligned} f(x) &\propto \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2} \\ \Rightarrow \log f(e^x) &= C - \frac{\nu+1}{2} \log\left(1 + \frac{1}{\nu}e^{2x}\right) \\ \Rightarrow \frac{d^2}{dx^2} \log f(e^x) &= \frac{d}{dx} - \frac{\nu+1}{2} \frac{\frac{2}{\nu}e^{2x}}{1 + \frac{1}{\nu}e^{2x}} = -\frac{\nu+1}{2} \frac{\frac{4}{\nu}e^{2x}}{(1 + \frac{1}{\nu}e^{2x})^2} < 0, \end{aligned}$$

and hence has log-concave $f(e^x)$.

Creating a symmetric distribution from the *Pareto* distribution, analogous to the double-exponential distribution, with $\alpha > 0$ gives

$$f(x) \propto x^{-\alpha-1} \Rightarrow \log f(e^x) = C - (\alpha+1)x,$$

with log-linear and hence log-concave $f(e^x)$.

6.3 Corollary 2

Proof. Normalize $\mathbb{E}[X] = \mathbb{E}[\tilde{X}] = 0$ (the general case follows by translation). $S = X + \varepsilon$ and $\tilde{S} = \tilde{X} + \varepsilon$ are location experiments with state ε and noise X and $\tilde{X}$, respectively. Since the density of ε is log-concave and $\tilde{X}$ is more precise than X , Theorem 1 implies that $\mathbb{E}[\varepsilon|\tilde{S} = s]$ is weakly further from $\mathbb{E}[\varepsilon] = 0$ than is $\mathbb{E}[\varepsilon|S = s]$. By $X = S - \varepsilon$, we have $\mathbb{E}[X|S = s] = s - \mathbb{E}[\varepsilon|S = s]$, so $\mathbb{E}[\tilde{X}|\tilde{S} = s]$ is weakly closer to the prior mean than is $\mathbb{E}[X|S = s]$. $\square$

6.4 Proposition 1

Proof. By symmetry and translation invariance of location experiments, it is without loss to suppose $x \geq \mathbb{E}[X] = 0$.

First, we show the inequality $\mathbb{E}[X] = 0 \leq \mathbb{E}[\mathbb{E}[X|S]|X = x]$. We have

$$\begin{aligned} \mathbb{E}[\mathbb{E}[X|S]|X = x] &= \int_{-\infty}^{\infty} \mathbb{E}[X|S = s] f_{\varepsilon}(s - x) ds \\ &= \int_0^{\infty} (\mathbb{E}[X|S = s] f_{\varepsilon}(s - x) + \mathbb{E}[X|S = -s] f_{\varepsilon}(-s - x)) ds. \end{aligned} \quad (10)$$

By symmetry of the prior and noise densities, $\mathbb{E}[X|S = s] = -\mathbb{E}[X|S = -s]$ and by Fact 1, $\mathbb{E}[X|S = s] > 0$. By symmetry around 0 and quasi-concavity of the noise density, $f_\varepsilon(-s - x) = f_\varepsilon(s + x) < f_\varepsilon(s - x)$ and hence, the integrand of (10) is positive for all $s \in [0, \infty)$.

Second, we show $\mathbb{E}[\mathbb{E}[X|S]|X = x] \leq x$. We have by symmetry of f_ε around 0,

$$\begin{aligned} x &= \int_{-\infty}^{\infty} s f_\varepsilon(s - x) ds \\ &= \int_0^{\infty} (s f_\varepsilon(s - x) + (-s) f_\varepsilon(-s - x)) ds. \end{aligned} \quad (11)$$

We need to show that (10) is less or equal to (11). By Fact 1, $s \geq \mathbb{E}[X|S = s]$ but $-s \leq \mathbb{E}[X|S = -s]$, preventing a direct conclusion of the result. However, by symmetry, $s - \mathbb{E}[X|S = s] = \mathbb{E}[X|S = -s] - (-s)$ and as before $f_\varepsilon(s - x) > f_\varepsilon(-s - x)$. Thus, for every $s \in [0, \infty)$, the integrand of (10) is less or equal to the integrand of (11). $\square$

6.5 Proposition 2

Proof. As in the proof of Proposition 1, it is without loss to suppose that $x > \mathbb{E}[X] = 0$.

The proof of the inequalities $x \geq \mathbb{E}[\mathbb{E}_A[X|S]|X = x]$ and $\mathbb{E}[\mathbb{E}_B[X|S]|X] \geq \mathbb{E}[X]$ is identical to the proof of Proposition 1.

It remains to show that $\mathbb{E}[\mathbb{E}_A[X|S]|X = x] \geq \mathbb{E}[\mathbb{E}_B[X|S]|X = x]$.

$$\begin{aligned} \mathbb{E}[\mathbb{E}_A[X|S]|X = x] &= \int_{-\infty}^{\infty} \mathbb{E}_A[X|S = s] f_\varepsilon(s - x) ds \\ &= \int_0^{\infty} (\mathbb{E}_A[X|S = s] f_\varepsilon(s - x) + \mathbb{E}_A[X|S = -s] f_\varepsilon(-s - x)) ds \\ \mathbb{E}[\mathbb{E}_B[X|S]|X = x] &= \int_{-\infty}^{\infty} \mathbb{E}_B[X|S = s] f_\varepsilon(s - x) ds \\ &= \int_0^{\infty} (\mathbb{E}_B[X|S = s] f_\varepsilon(s - x) + \mathbb{E}_B[X|S = -s] f_\varepsilon(-s - x)) ds \\ \Rightarrow \mathbb{E}[\mathbb{E}_A[X|S]|X = x] - \mathbb{E}[\mathbb{E}_B[X|S]|X = x] &= \\ &= \int_0^{\infty} ((\mathbb{E}_A[X|S = s] - \mathbb{E}_B[X|S = s]) f_\varepsilon(s - x) + (\mathbb{E}_A[X|S = -s] - \mathbb{E}_B[X|S = -s]) f_\varepsilon(-s - x)) ds \\ &= \int_0^{\infty} \left(\underbrace{(\mathbb{E}_A[X|S = s] - \mathbb{E}_B[X|S = s])}_{\geq 0} \underbrace{(f_\varepsilon(s - x) - f_\varepsilon(-s - x))}_{\geq 0} \right) ds \geq 0 \end{aligned}$$

From the second-to-last to the last line we used antisymmetry of the conditional-expectation functions $\mathbb{E}_A[X|S = s]$ and $\mathbb{E}_B[X|S = s]$ in s , which are due to symmetry of the prior and the noise density. In the last line, we used Theorem 1, by which $\mathbb{E}_A[X|S = s] \geq \mathbb{E}_B[X|S = s]$ and we used symmetry and quasi-concavity of the noise density, by which $f_\varepsilon(s - x) \geq f_\varepsilon(s + x) = f_\varepsilon(-s - x)$ under $s, x \geq 0$. $\square$

6.6 Proposition 3

The proof of Proposition 3 is analogous to the proof of Proposition 2.